

School Management System - Class Management Module

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-- School Management System - Class Management Module Schema

CREATE TABLE classes (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,
    class_code VARCHAR(20) UNIQUE,
    class_name VARCHAR(50),
    capacity INT,
    academic_year VARCHAR(20),
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);

CREATE TABLE sections (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,
    section_name VARCHAR(20),
    class_id BIGINT,
    class_teacher BIGINT NULL,
    created_at TIMESTAMP,
    updated_at TIMESTAMP,
    FOREIGN KEY (class_id) REFERENCES classes(id) ON DELETE CASCADE
);

CREATE TABLE subjects (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,
    subject_code VARCHAR(20) UNIQUE,
    subject_name VARCHAR(100),
    is_optional BOOLEAN DEFAULT 0,
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);

CREATE TABLE class_subjects (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,
    class_id BIGINT,
    subject_id BIGINT,
    teacher_id BIGINT NULL,
    created_at TIMESTAMP,
    updated_at TIMESTAMP,
    FOREIGN KEY (class_id) REFERENCES classes(id) ON DELETE CASCADE,
    FOREIGN KEY (subject_id) REFERENCES subjects(id) ON DELETE CASCADE
);

CREATE TABLE students (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,
    student_id VARCHAR(20) UNIQUE,
    name VARCHAR(100),
    class_id BIGINT,
    section_id BIGINT,
    roll_number INT,
    admission_date DATE,
    created_at TIMESTAMP,
    updated_at TIMESTAMP,
    FOREIGN KEY (class_id) REFERENCES classes(id),
    FOREIGN KEY (section_id) REFERENCES sections(id)
);

CREATE TABLE timetables (
    id BIGINT AUTO_INCREMENT PRIMARY KEY,
    class_id BIGINT,
    section_id BIGINT,
    subject_id BIGINT,
    teacher_id BIGINT,
    day_of_week ENUM('Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat'),
    period_number INT,
    start_time TIME,
    end_time TIME,
    FOREIGN KEY (class_id) REFERENCES classes(id),
    FOREIGN KEY (section_id) REFERENCES sections(id),
    FOREIGN KEY (subject_id) REFERENCES subjects(id),
    FOREIGN KEY (teacher_id) REFERENCES teachers(id)
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);  
  
CREATE TABLE teachers (  
    id BIGINT AUTO_INCREMENT PRIMARY KEY,  
    teacher_id VARCHAR(20) UNIQUE,  
    name VARCHAR(100),  
    email VARCHAR(100) UNIQUE,  
    phone VARCHAR(20),  
    specialization VARCHAR(100),  
    created_at TIMESTAMP,  
    updated_at TIMESTAMP  
);
```